

Evangelia (Eva) Deliporanidou

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Summary

PhD candidate in Astrophysics at the University of Cambridge with advanced programming, mathematical, and data analysis skills. Experienced in statistical modelling, machine learning, and problem-solving in complex scientific datasets.

Education

- University of Cambridge**, PhD in Astrophysics Oct 2023 – Present
- Design and refine algorithms to predict solar and stellar flux variation
 - Applying regression techniques, statistical modelling and ML to interpret astrophysical trends
- University College London (UCL)**, MSc in Astrophysics Sept 2022 – Sept 2023
- **MSc Thesis:** Towards a 3D map of the dust of the Nuclear Stellar Disk of the Milky Way
- University of Strathclyde**, BSc (Hons) in Physics Sept 2019 – Sept 2022
- First Class Honours (80/100) and deferred entry into second year

Skills & Tools

- Coding Languages:** Python, IDL, R (Beginner), Matlab (Intermediate), SQL (Intermediate), Fortran (Intermediate), HTML (Intermediate).
- Libraries:** numpy, pandas, PyTorch, TensorFlow, scikit-learn.
- Machine Learning & AI:** Deep Learning, Physics-informed ML, Bayesian Statistics, Markov Chain Monte Carlo.
- Languages:** Greek (Native), English (Fluent), German (Intermediate), Spanish (Beginner).

Experience

- Undergraduates' Supervisor**, University of Cambridge Oct 2022 – Present
- Guiding students through complex differential equations, Lagrangian & Hamiltonian mechanics and breaking down abstract mathematical problems
- Fundraiser**, Buffalo Fundraising Consultants Oct 2021 - Oct 2022
- Optimising call strategies, improving engagement efficiency
 - Handling approximately 60 dials / 2 full calls per hour with a total success rate of 68 %

Additional Education/Experience

- QRT Trading Academy**, Cambridge University Algorithm Trading Society (CUATS) 2026
- Trading & Risk Management
 - Alpha modelling, portfolio construction, execution in practise
 - LSEG data training and use
- Machine Learning**, DeepLearning.AI - Stanford (via Coursera) 2025
- Supervised ML: Regression & Classification
 - Advanced Learning Algorithms
 - Unsupervised Learning, Recommenders, Reinforcement Learning
- Data Analysis Workshop**, Imperial College London 2024
- Learning Bayesian methods and numerical techniques for astrophysical data analysis.

- Applying data analysis approaches to large datasets, improving model selection and parameter estimation.

CS50's Introduction to Cybersecurity, Harvard (online) 2024

- Learning core cybersecurity principles: threat identification, risk assessment, and mitigation.
- Gaining hands-on skills in account security, multi-factor authentication, and social engineering awareness.

Publications

CHIANTI—An Atomic Database for Emission Lines—Paper. XVIII. Version 11, Advanced Ionization Equilibrium Models: Density and Charge Transfer Effects Oct 2024

R.P. Dufresne, Giulio Del Zanna, P. R. Young, K. P. Dere, Evangelia Deliporanidou, W. T. Barnes, E. Landi
[10.3847/1538-4357/ad6765](https://doi.org/10.3847/1538-4357/ad6765)

Modelling Stellar Irradiances I: The transition regions of FGKM stars Oct 2024

Evangelia Deliporanidou, Giulio Del Zanna
[10.1093/mnras/stae2299](https://doi.org/10.1093/mnras/stae2299)

H, He-like recombination spectra VI: quadrupole l-changing collisions Apr 2025

Evangelia Deliporanidou, N R Badnell, P J Storey, G Del Zanna, G J Ferland,
[10.1093/mnras/staf608](https://doi.org/10.1093/mnras/staf608)

Modelling Stellar Irradiances – II. Correlations of solar irradiances with proxies of activity along a cycle May 2025

Evangelia Deliporanidou, G Del Zanna, T N Woods, D Woodraska
[10.1093/mnras/staf812](https://doi.org/10.1093/mnras/staf812)

Modelling Stellar Irradiance - III: A predictive model for solar irradiances Feb 2026

Evangelia Deliporanidou, G Del Zanna
[10.1093/mnras/stag412](https://doi.org/10.1093/mnras/stag412)

SIRIUS: stellar and ISM research via in-orbit ultraviolet spectroscopy Aug 2026

Barstow et al., Deliporanidou E.
[10.1117/12.3104810](https://doi.org/10.1117/12.3104810)

Awards & Honours

PhD stipend: Modelling the emission of the outer solar atmosphere

Talks & Conference Participation

- **Cool Stars 22:** Modelling stellar TR of FGKM stars with advanced ionisation equilibrium
- **NAM 2024, ESPM-17:** Correlation of solar irradiances with proxies of activity along a cycle
- **UKSP, ESPOS, NAM 2025** A predictive model for solar irradiances

Societies & Activities

- President of the Cambridge University Hellenic Society (2025/26)
- Secretary of the Cambridge University Hellenic Society (2024/25)
- Member of the Cambridge University Algorithmic Trading Society
- Fundraised over £700 for Alzheimer's Research UK through participation in the Cambridge Half Marathon (2026)